

Graziano Montanaro

Data Engineer | Distributed Systems & Platform

Italy · EU citizen · Open to relocation/remote

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Experience

- 09/2024– Present **Data Engineer, Intella, Milan**
- Set the company's data engineering standards and built internal Python libraries adopted by data, data-science, and full-stack engineers, standardizing pipeline development and quality checks.
 - Re-architected ETL processes: migrated TB-scale datasets from a legacy storage format to Delta Lake, rebuilt batch orchestration with Prefect, and added data quality checks, improving reliability of data assets. Cut storage costs and improved query performance by up to 90%.
 - Developed a microservice orchestrating hundreds of concurrent jobs, meeting SLA requirements with fault tolerance and efficient resource utilization.
- 02/2023– 09/2024 **Data Engineer, Tuidi, Putignano**
- Optimized ETL pipelines integrating multiple data sources over TB-scale datasets, combining distributed PySpark on Databricks with single-node Polars for smaller workloads, cutting processing time by up to 70%.
 - Introduced data quality checks and monitoring, improving data reliability and downstream ML model accuracy.
 - Built a model-monitoring dashboard in Dash for prediction evaluation and detection of performance degradation and target shift.

Education

- 09/2020– 07/2022 **Master's Degree in Computer Science and Artificial Intelligence, University of Bari Aldo Moro, Bari, 110/110 cum laude**
- Thesis on multimodal 3D CNN for dementia detection using MRI and amyloid PET scans. Achieved state of the art results on the OASIS-3 cohort (95% accuracy). Published in Scientific Reports in 2024. Used XAI techniques (SHAP, Grad-CAM) to provide explainability and validate the results obtained.
- 09/2017– 07/2020 **Bachelor's Degree in Computer Science, University of Bari Aldo Moro, Bari, 110/110 cum laude**
- Developed a web platform in Laravel for performing static analysis of web pages to discover UX smells before usability tests with users.

Skills

Programming	Python, Java, SQL
Data & Lakehouse	PySpark, Databricks, Delta Lake, Iceberg, dbt, Airflow, Prefect, Pandas, Polars, DuckDB, Ibis, Great Expectations
Streaming & Databases	Kafka, RabbitMQ, PostgreSQL, MySQL, MongoDB, Redis
Cloud & DevOps	AWS (S3, Glue), Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Grafana, Git, DVC
Languages	English (Fluent), Italian (Native)

Projects

Italy Dashboard, [GitHub](#)

Built a snapshot-first data platform unifying official Italian statistics (ISTAT, INPS, MUR, Open-Meteo/Copernicus) for public-source verification.

- Designed dbt + DuckDB + Parquet pipelines in Python/Polars, serving two independent frontends (Reflex, TypeScript/DuckDB-WASM) with CI-enforced conformance and data-provenance checks.

Poor Man Lakehouse, [GitHub](#)

Architected a composable, open-source lakehouse from scratch.

- Designed a pluggable layer supporting multiple catalogs, query engines, and table formats.

Publications

- 2024 **Automated detection of Alzheimer's disease: a multi-modal approach with 3D MRI and amyloid PET.** Giovanna Castellano, Andrea Esposito, Eufemia Lella, **Graziano Montanaro**, Gennaro Vessio. *Sci Rep*, 14, 5210.
- 2021 **Detection of Dementia through 3D convolutional neural networks based on amyloid PET.** Giovanna Castellano, Andrea Esposito, Eufemia Lella, **Graziano Montanaro**, Gennaro Vessio. *2021 IEEE SSCI*.